

COMPUTER VISION

The Cameras Are Already There

A deployment guide for PPE and safety monitoring on the plant CCTV you already own.

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SUMMARY

Most plants own 30 to 120 cameras that only get watched after an incident. Here is the honest path from that footage to a live safety signal, including the maths that decides which cameras can actually see a hard hat.

Walk into almost any factory in Harare, Bulawayo, or Mutare and you will find a wall of monitors showing thirty camera feeds. The cameras were bought for insurance, incident review, and the occasional dispute about a missing pallet. They record continuously, nobody watches them, and the footage is opened only after something has already gone wrong.

The premise of this paper is that the expensive part already exists. Turning those cameras into a live safety signal is mostly a software, geometry, and process problem - and the geometry is where most projects quietly fail.

What commodity CCTV actually gives you

Spec	Typical installed base	What it means for detection
Resolution	2 MP (1920 x 1080)	Enough for people at 20 m; not enough for helmet detail at that range.
Frame rate	15-25 fps	Fine for walking pace; marginal for fast-moving forklifts.
Lens	2.8-12 mm varifocal, often fixed wide	Wide lenses buy coverage and lose pixels per metre.
Stream	H.264 / H.265 over RTSP, ONVIF	Workable. You need the sub-stream for analysis and the main stream for evidence.
Night	IR illuminators, monochrome	Colour-based rules stop working; vests still resolve, dark helmets do not.
Placement	3-5 m ceiling, angled down	Good for zones and counting, hard for face-on PPE checks.

The pixel maths that decides the project

Before anyone trains a model, measure. A standard hard hat is roughly 25 to 30 centimetres across. To detect it reliably you want at least 60 to 80 pixels across that width in the image. On a 1080p sensor, 80 pixels is about 4 per cent of the frame width, which means the helmet must occupy less than a twenty-fifth of the horizontal field of view.

In practice: a 1080p camera with a 10 m wide field of view gives roughly 190 pixels per metre, so a helmet is around 50 pixels wide - usable. The same camera covering a 30 m warehouse aisle gives about 64 pixels per metre, so a helmet is 18 pixels wide - not usable, no matter which model you buy. You will detect the person and guess at the hat.

Run this before the quote

For every camera, write down the horizontal width it covers at the distance where people actually walk. Divide 1920 by that width. Multiply by 0.28 m. If the answer is under 50 pixels, that camera is a counting camera, not a PPE camera.

Six failure modes worth naming early



A fixed ceiling camera over a production walkway. A hard hat resolves at this distance; a glove does not.

— Distance. The most common cause of a failed pilot, and the easiest to predict with the maths above.

— Backlight and glare. Roller doors and high windows blow out vests into white rectangles at exactly the times of day when the shift changes.

— Occlusion. Racking, forklifts, and columns hide the torso for seconds at a time. A single-frame model sees a violation where a tracker sees a person walking behind a rack.

— Night and IR. Monochrome footage removes the colour cue that makes a hi-vis vest obvious, and IR hotspots wash out reflective strips.

— Colour and contrast. Navy overalls against a dark floor, or a dark green vest against a green machine, will produce confident nonsense.

— Angle. A ceiling camera looking down sees the top of a helmet and almost none of the face, gloves, or eyewear. Do not promise eye protection detection from a 60-degree downward view.

Step 1: a camera audit that produces a decision

Pull ten frames from each camera at three times of day - early shift, midday, night with lights on. Score each camera for the specific use case, not in general. A camera can be excellent for zone intrusion and useless for helmet checks. The output should be a table somebody can act on.

Verdict	Meaning	What to do
Green	Helmet resolves above 60 px; lighting stable; a person is visible for 3+ seconds	Include in the pilot
Amber	Person resolves, helmet marginal; or frequent occlusion	Use for zones, counting, or presence - not PPE
Red	Under 40 px, heavy glare, or lens damaged	Re-aim, add a camera, or leave it out

Step 2: two weeks of your own footage

Public datasets get you a starting model; they do not get you a model that works on your floor. The difference is your lighting, your uniform colours, your camera angle, and your definition of a violation. Collect 1,500 to 3,000 labelled frames per camera-condition pair, sampled across shifts, and deliberately include the hard cases: night, glare, crowds, partial occlusion, people carrying boxes.

Label discipline matters more than model choice. Define classes precisely - person, hard hat, safety vest, and optionally gloves and eyewear - and write down what counts as worn. A helmet hanging from a belt is not worn. A vest over a shoulder is not worn. If two people label the same 100 frames and disagree on more than five, the definition is not ready.

Step 3: evaluate like an operator, not a researcher

Mean average precision is a fine training signal and a poor operations metric. What the plant cares about is: how many real violations did we catch, and how many alerts did we send that were wrong? Those are recall at your chosen threshold and false alerts per shift.

Two mechanisms do most of the work of turning frame-level predictions into tolerable alerts. The first is persistence: a violation must be visible for two to four seconds across several frames before it counts. The second is identity: attach detections to a track, so one person walking past a camera produces one alert, not forty. Zone gating - only count violations inside a defined polygon - removes the corridor, the car park, and the reflection in the window.

> **A realistic target for a first deployment**
>
> Recall above 85 per cent on the site's own validation set, with fewer than three false alerts per camera per shift. If a vendor quotes 99 per cent without showing you their confusion matrix on your footage, ask what the denominator was.

Step 4: design the alert before the model

Most failed safety deployments are not model failures; they are notification failures. A stream of alerts into a group chat is muted within a week. Decide, in writing, who receives what, at what hour, and what they are expected to do about it.

— Immediate alert to the shift supervisor for a person in a restricted zone, with a ten-second clip and the camera name.

— Batched digest to the safety officer every morning for PPE violations, with counts by line and shift - not one message per event.

— No alerts to the executive team. They get a weekly number with a trend line and the three worst cameras.

— Every alert carries evidence: timestamp, camera, clip, and the detection confidence. An alert without evidence gets argued about instead of acted on.

Step 5: where the inference runs

In Southern Africa, the honest answer is usually on-premises. Load-shedding interrupts cloud connectivity more often than it interrupts a UPS-backed switch, and a safety system that stops working during a power event is not a safety system. A small edge box per site - Jetson-class, fanless, on a UPS - can run several camera streams locally and forward only events and short clips.

Option	Strengths	Watch out for
Edge box on site	Works through WAN outages; video stays on site; predictable latency	Capital cost; physical security; remote updates need a plan
Plant server / VM	Cheaper if you already have GPU capacity; central management	Shared with other workloads; no UPS isolation
Cloud inference	No hardware; easy to scale across sites	Bandwidth cost and load-shedding fragility; data residency questions

Privacy, law, and the conversation nobody wants to have

Workplace cameras are normal. Workplace cameras that classify people are a different conversation, and it should happen before the pilot, not after a rumour. Zimbabwe's Cyber and Data Protection Act [Chapter 12:07] (2021) applies to personal data processed by cameras, including images. Expect to need a lawful basis, a defined purpose, minimum necessary collection, a retention limit, and appropriate security.

Our default design for this kind of system is deliberately conservative: no facial recognition, no identification of individuals, no audio, alerts aggregated by zone and shift rather than attached to a person's name, and raw video not stored at all where inference happens at the edge. Signage at entrances, a written notice to workers and their representatives, and a named contact for questions. This is not legal advice - take local counsel on your obligations - but it is the difference between a deployment that lasts and one that gets switched off.

Timeline and budget shape

Phase	Duration	Output
Camera audit	3-5 days	Per-camera verdict table and a go/no-go on the use case
Dataset and labelling	2 weeks	Site-specific labelled set, class definitions, held-out validation set
Model training and tuning	1-2 weeks	Model card with per-class metrics on your footage
Workflow, alerts, integration	1 week	WhatsApp/email alerts, dashboard, clip retention
Shadow running	2 weeks	Alerts reviewed daily against operator judgement; false-alert tuning
Handover and review	1 week	Runbook, threshold settings, monthly review rhythm

A single-camera, single-use-case pilot is a four to six week project. Multi-site rollouts add integration and change management, not modelling time.

What this approach will not do

It will not see through a forklift, resolve a helmet at forty metres, or tell you whether gloves are appropriate for the task. It will not replace a safety officer's judgement, and it will not fix a culture where violations are tolerated. It gives the safety team a second pair of eyes that never blinks and never gets tired - which is useful, and only useful if somebody acts on what those eyes see.

A NOTE ON HONESTY

We would rather lose a sale than deploy a camera that cannot see the thing it is being asked to see. If the audit says five of your forty cameras are suitable, the pilot runs on five cameras.

ABOUT THIS PAPER

CloudyAI is a Zimbabwean AI studio. We deploy computer vision on existing plant cameras, audit AI spend, and help teams respond when an AI system misbehaves. Research is published free, without an email wall.

Corrections and questions: research@cloudyai.co.zw. If you would like this method applied to your plant, the first conversation is free and the camera audit is a fixed-price week.